

SAP CPI Project Documentation

Author: Rohitha

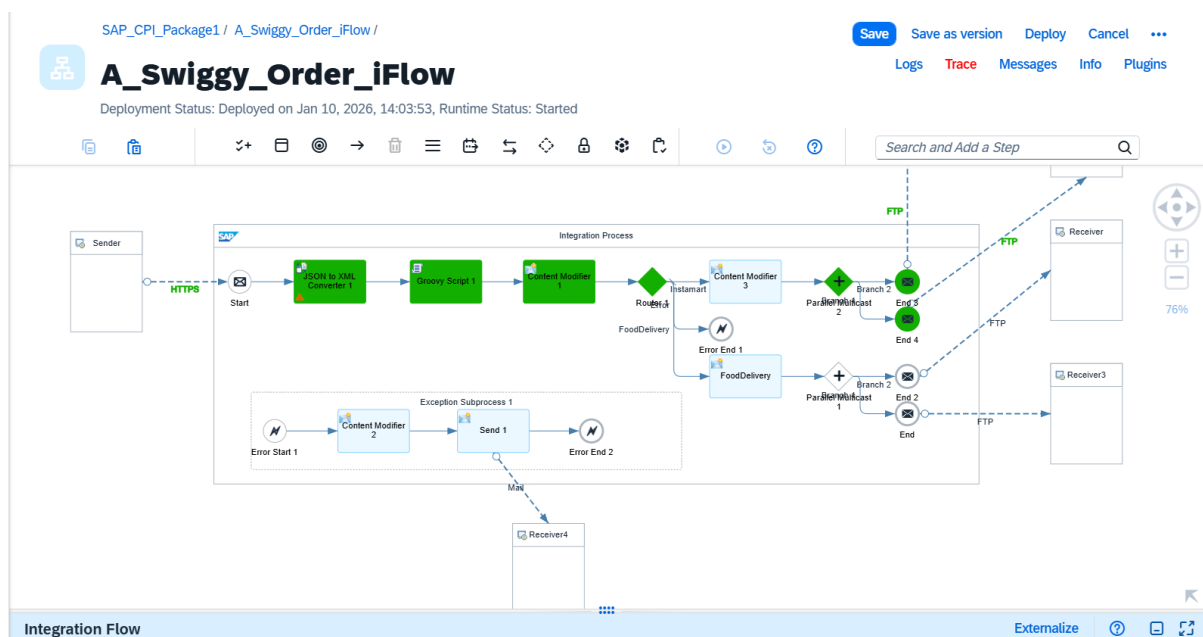
Date:29/12/2025

Scenario: Multi-Channel Order Routing Integration

1.Scenario Overview

This SAP CPI integration scenario demonstrates a real-world order processing use case where customer orders are received via a REST API in JSON format, validated through Groovy scripting, routed based on order type (Instamart or FoodDelivery), and handled accordingly. Success and failure flows are implemented with proper error messaging, exception subprocess and success confirmation

2.Architechture Diagram



3. Step-by-Step Implementation

The Integration flow includes the following main steps:

- HTTPS Sender Adapter to receive incoming orders in JSON format
- JSON to XML Converter used to convert input JSON to XML format
- Groovy Script to validate mandatory fields and throw exception with custom error message

- Router to branch based on Order type (Instamart vs FoodDelivery)
- Success branch: Set response using Content Modifier
- Failure handled by Exception Subprocess which formats and sends error response email
- Message Logging enabled for traceability

4.Adapter Configuration:

HTTPS Sender Adapter

Endpoint: *https://b84a8864trial.it-cpitrial05-rt.cfapps.us10-001.hana.ondemand.com/http/FoodOrder*

Method: POST

Authentication: Basic

HTTPS

General **Connection** Conditions

REQUEST PROCESSING

Address: /FoodOrder

Authorization: User Role

User Role: ESBMessaging.send

CSRF Protected: ☒

FTP Receiver Adapter:

FTP

General **Target** Processing

FILE ACCESS PARAMETERS

Directory: Instamart

File Name: Groceries_Order

Append Timestamp: ☒

CONNECTION PARAMETERS

Address: ftp.drivehq.com

Proxy Type: Internet

Encryption: Explicit FTPS

Credential Name: FTP_Connect2

Timeout (in ms): 10000

Maximum Reconnect Attempts: 3

Reconnect Delay (in ms): 1000

5.Integration Components:

JSON_to_XML_Converter :

Adding a root element Orders to input

JSON To XML Converter

General

Processing

Use Namespace Mapping:

☒

Namespace Mapping:

JSON Prefix	XML Namespace

JSON Prefix Separator:

Colon(:)

Add XML Root Element:

☒

Name:

Orders

Groovy: CustomerOrderValidation:

If validation passes →

- Message body still contains the original XML
- Content Modifier can use it directly
- Properties available in Content Modifier:
 \${property.Type} which takes value from xpath
 \${property.Message Payload} retains input data

✗ If validation fails →

iFlow stops and throws exception and calls Exception Subprocess

Content Modifier:

Content Modifier

General

Message Header

Exchange Property

Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	Type	XPath	//OrderType	java.lang.String	
Create	Message Payload	Expression	\$(in.body)		

Router: Condition:

If OrderType = "FoodDelivery" → Route to "FoodDelivery" branch,
or OrderType = "Instamart" → Route to "Instamart" branch,
or any other OrderType → throws exception.

Router

General

Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Instamart	\$(property.Type) = 'Instamart'	☐
2	Error		☒
3	FoodDelivery	\$(property.Type) = 'FoodDelivery'	☐

6.Exception Subprocess: Handles errors and log details.

Alert [A_Swiggy_Order_iFlow] has caught an exception Inbox x



rohitham2019@gmail.com

to me ▼

Hello Team,

Message has been failed with below error:

Error detail: java.lang.Exception: java.lang.Exception: Missing mandatory field: OrderID@ line 10 in script4.groovy

Correlation ID: AGliDiERwq4zdX5NBGge0PvrDQEV

Message ID: AGliDiGaYKgpvdIGZwpTy0yASZ3V

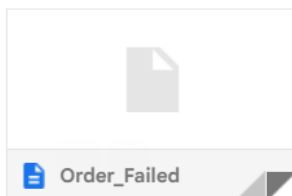
Iflow Name: A_Swiggy_Order_iFlow

Date: 10-01-2026 08:30 / SAP CPI - Tenant Number:

Thanks & Regards,

SAP CPI Team

One attachment • Scanned by Gmail ⓘ Add to Drive



7. Groovy Script:

```

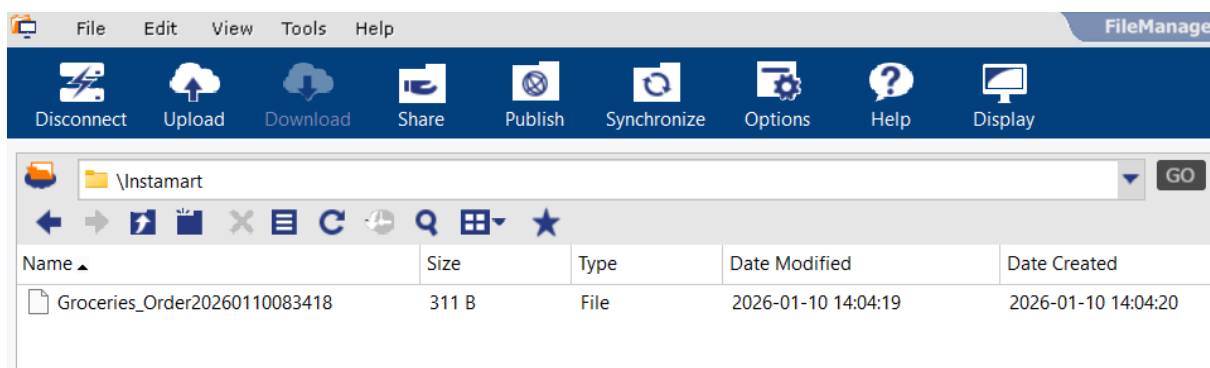
4 Message processData(Message message) {
5
6     String xml = message.getBody(String)
7
8     // ----- Order level validation -----
9     if (!xml.contains("<OrderID>") || xml.contains("<OrderID></OrderID>")) {
10         throw new Exception("Missing mandatory field: OrderID")
11     }
12
13     if (!xml.contains("<CustomerName>") || xml.contains("<CustomerName></CustomerName>")) {
14         throw new Exception("Missing mandatory field: CustomerName")
15     }
16
17     if (!xml.contains("<OrderType>") || xml.contains("<OrderType></OrderType>")) {
18         throw new Exception("Missing mandatory field: OrderType")
19     }
20
21     // ----- Item existence validation -----
22     if (!xml.contains("<Item>")) {
23         throw new Exception("Missing mandatory section: Item")
24     }
25
26     // ----- Item level validation -----
27     Pattern itemPattern = Pattern.compile("<Item>(.*?)</Item>", Pattern.DOTALL)
28     Matcher matcher = itemPattern.matcher(xml)
29
30     int itemCount = 0
31     while (matcher.find()) {
32         itemCount++
33         String itemXml = matcher.group(1)
34
35         if (!itemXml.contains("<Name>") || itemXml.contains("<Name></Name>")) {
36             throw new Exception("Missing mandatory field: Item Name at position " + itemCount)
37         }
38
39         if (!itemXml.contains("<Qty>") || itemXml.contains("<Qty></Qty>")) {
40             throw new Exception("Missing mandatory field: Item Qty at position " + itemCount)
41         }
42     }
43
44     message.setBody(xml)
45     return message
46 }
47

```

8. Input JSON:

```
1  {
2  "Order":
3  {
4      "OrderID": "ORD1A12",
5      "CustomerName": "Rubina",
6      "OrderType": "FoodDelivery",
7      "Items":
8      {
9          "Item":
10         [
11             {
12                 "Name": "Sandwich",
13                 "Qty": 2
14             },
15             {
16                 "Name": "PIZZA",
17                 "Qty": 2
18             },
19             {
20                 "Name": "Coffee",
21                 "Qty": 2
22             }
23         ]
24     }
25 }
26 }
```

9. Output : Order Placed in FTP server:



The screenshot shows the FileManager application interface. The top menu bar includes File, Edit, View, Tools, and Help. Below the menu is a toolbar with icons for Disconnect, Upload, Download, Share, Publish, Synchronize, Options, Help, and Display. The main area displays a file explorer view for the directory \Instamart. A table lists the files and folders in this directory.

Name	Size	Type	Date Modified	Date Created
Groceries_Order20260110083418	311 B	File	2026-01-10 14:04:19	2026-01-10 14:04:20

Order Processing

```
<?xml version='1.0' encoding='UTF-8'?>
<Orders>
  <Order>
    <OrderID>ORD1A12</OrderID>
    <CustomerName>Rubina</CustomerName>
    <OrderType>FoodDelivery</OrderType>
    <Items>
      <Item>
        <Name>Sandwich</Name>
        <Qty>2</Qty>
      </Item>
      <Item>
        <Name>PIZZA</Name>
        <Qty>2</Qty>
      </Item>
      <Item>
        <Name>Coffee</Name>
        <Qty>2</Qty>
      </Item>
    </Items>
  </Order>
</Orders>
```